

Visual System

Companion Figures

22 figures referenced from the lecture .md

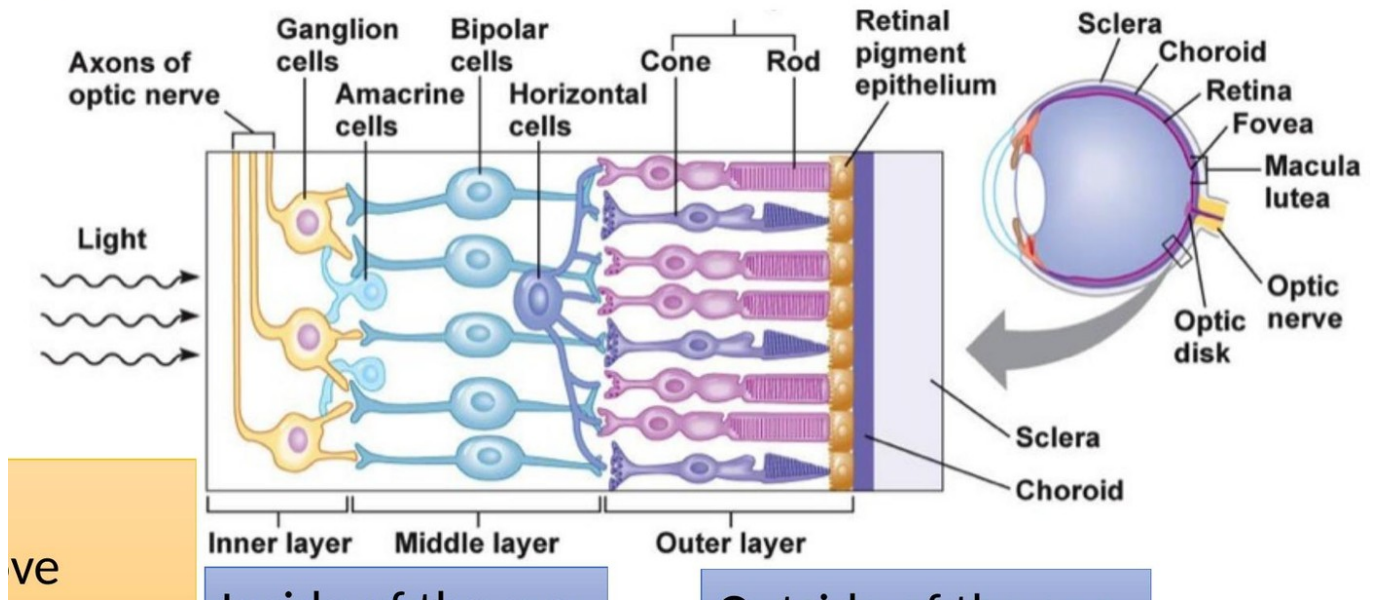


Figure L1.1 — Retinal layered structure: photoreceptors → bipolar → ganglion cells

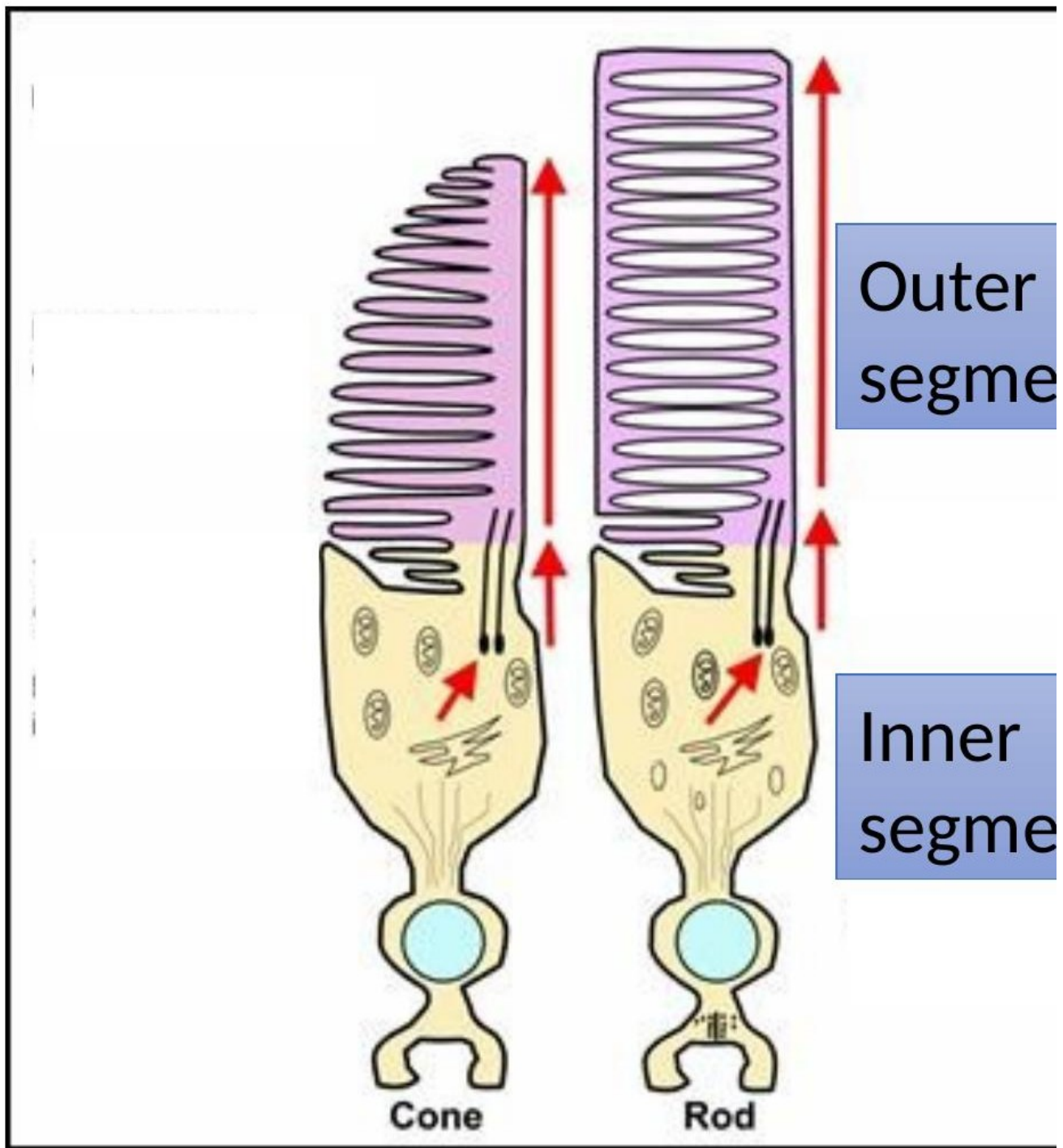


Figure L1.2 — Rod and cone photoreceptor structure

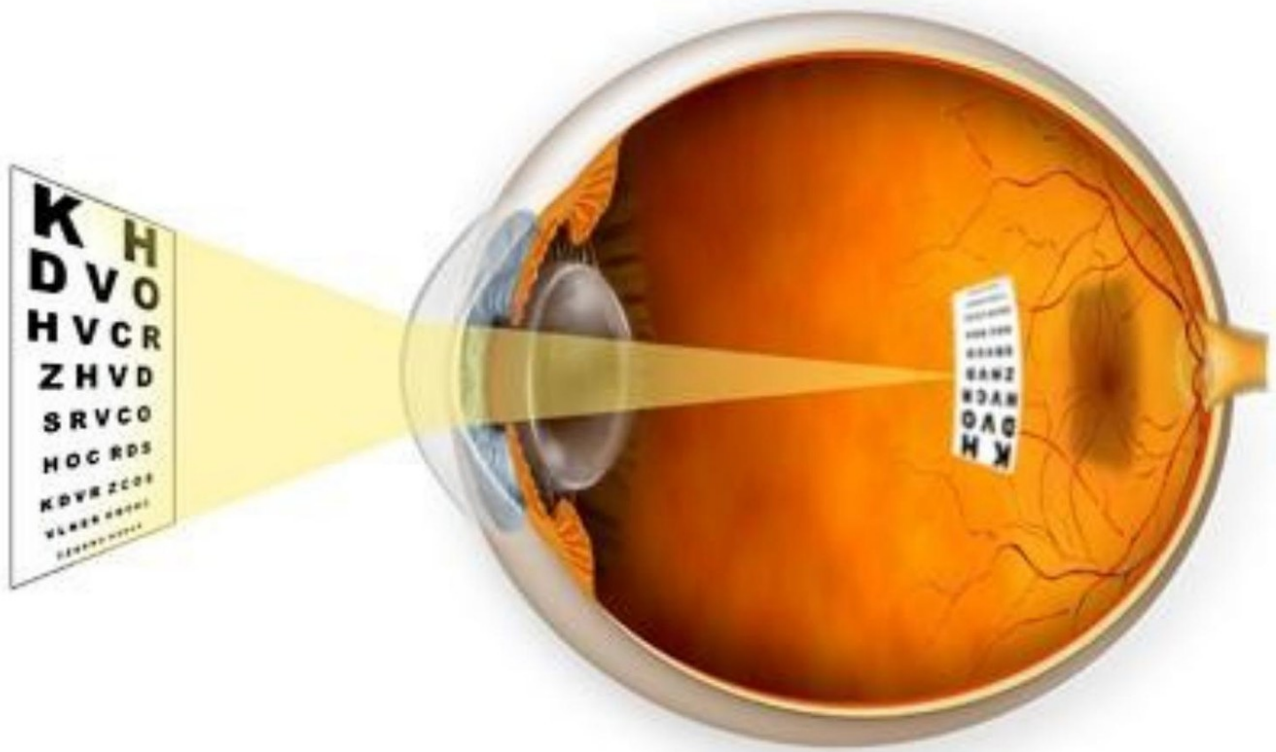


Figure L1.3 — Myopia: focus anterior to retina, long globe

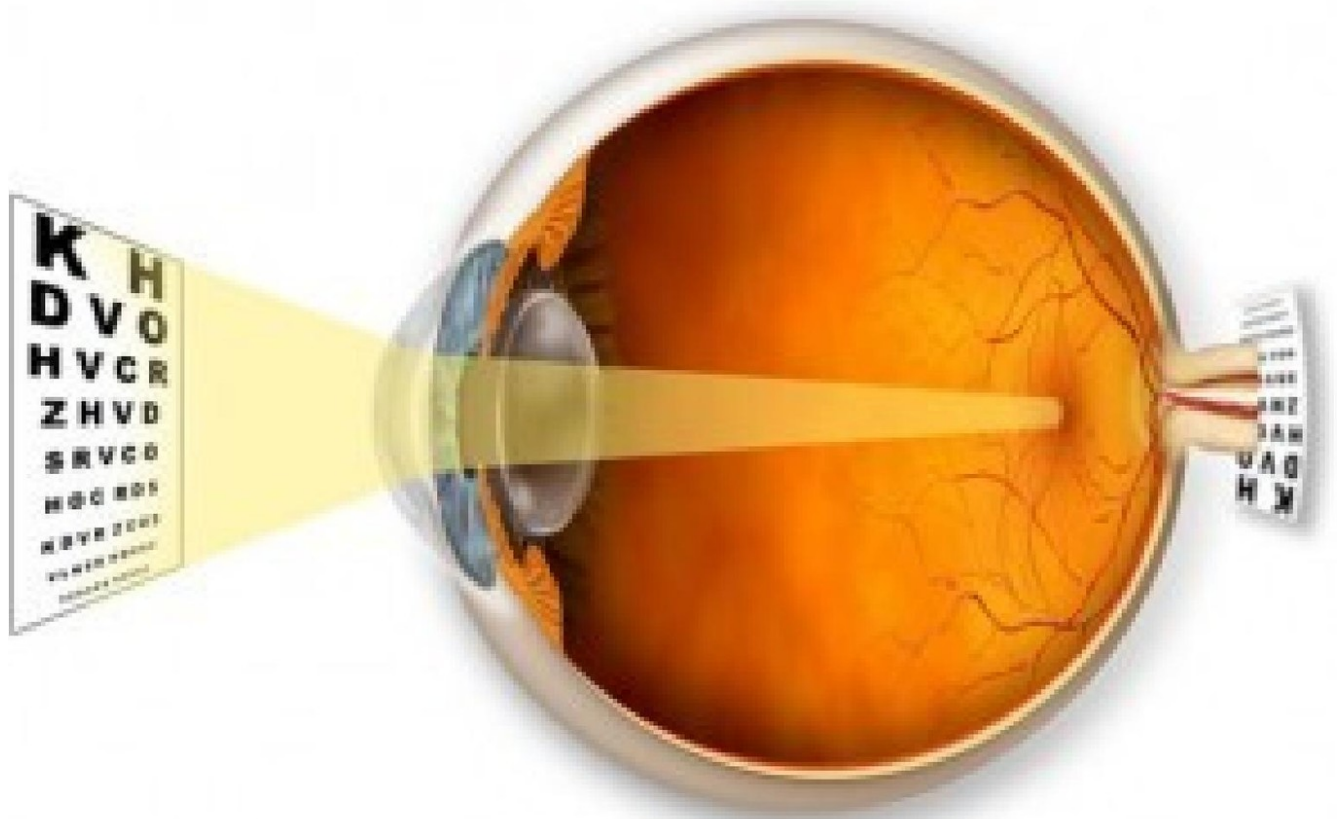


Figure L1.4 — Hyperopia: focus posterior to retina, short globe

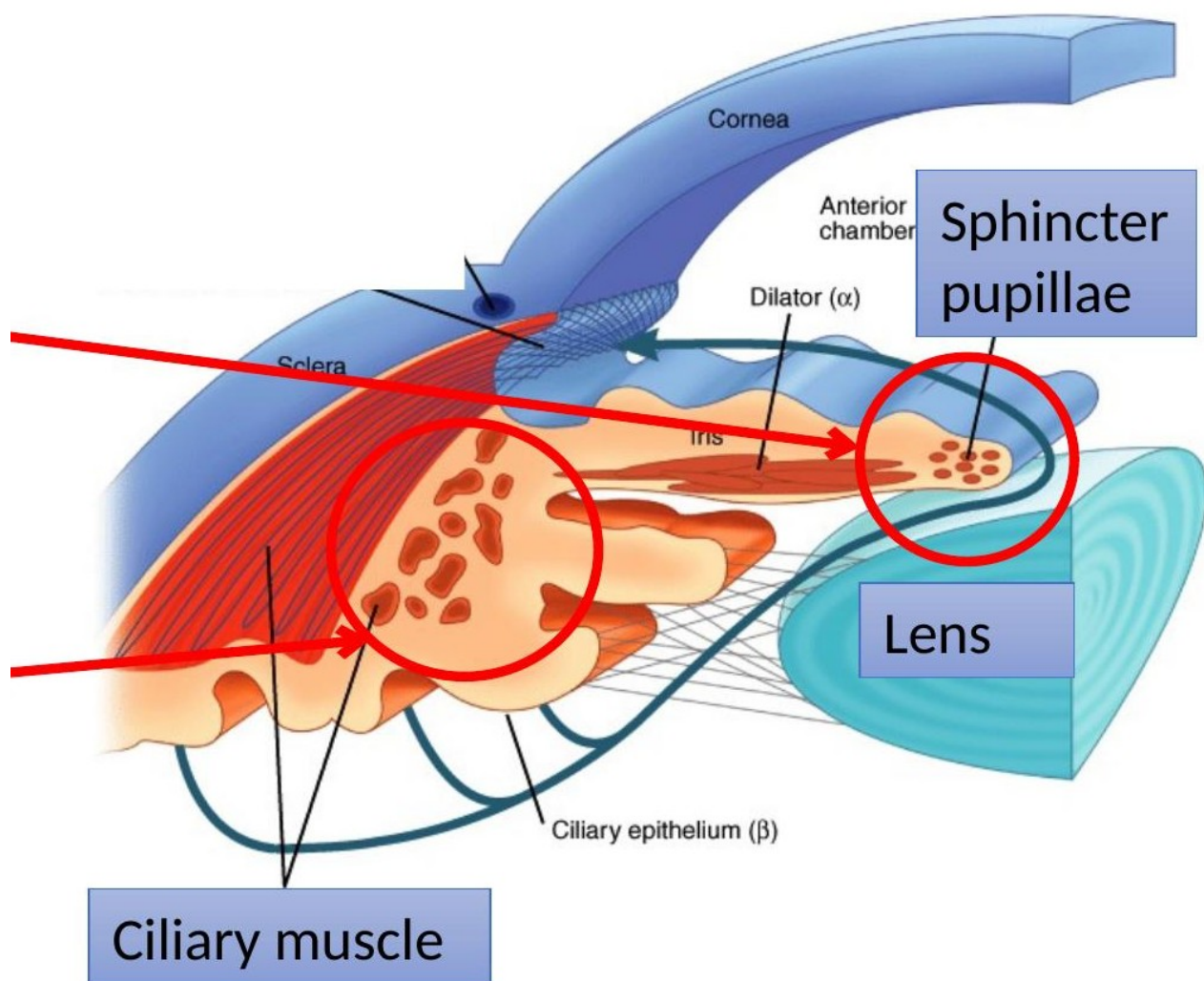


Figure L1.5 — Near response triad anatomy

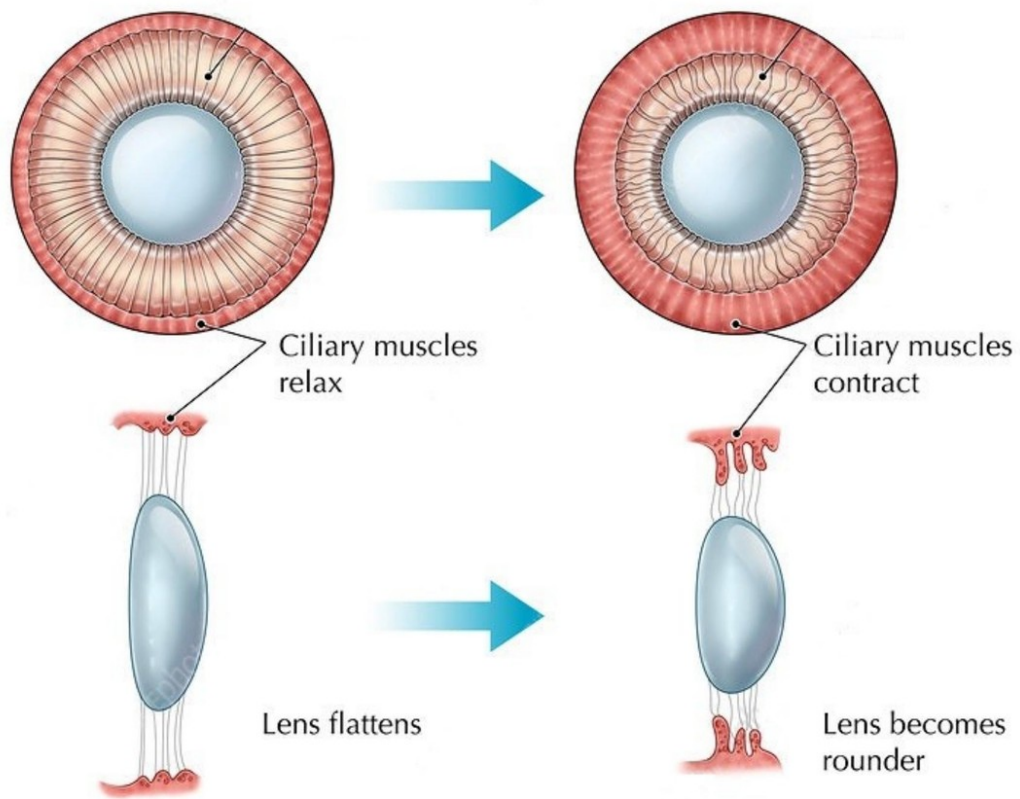


Figure L1.6 — Accommodation: ciliary muscle relax (flat lens) vs contract (round lens)

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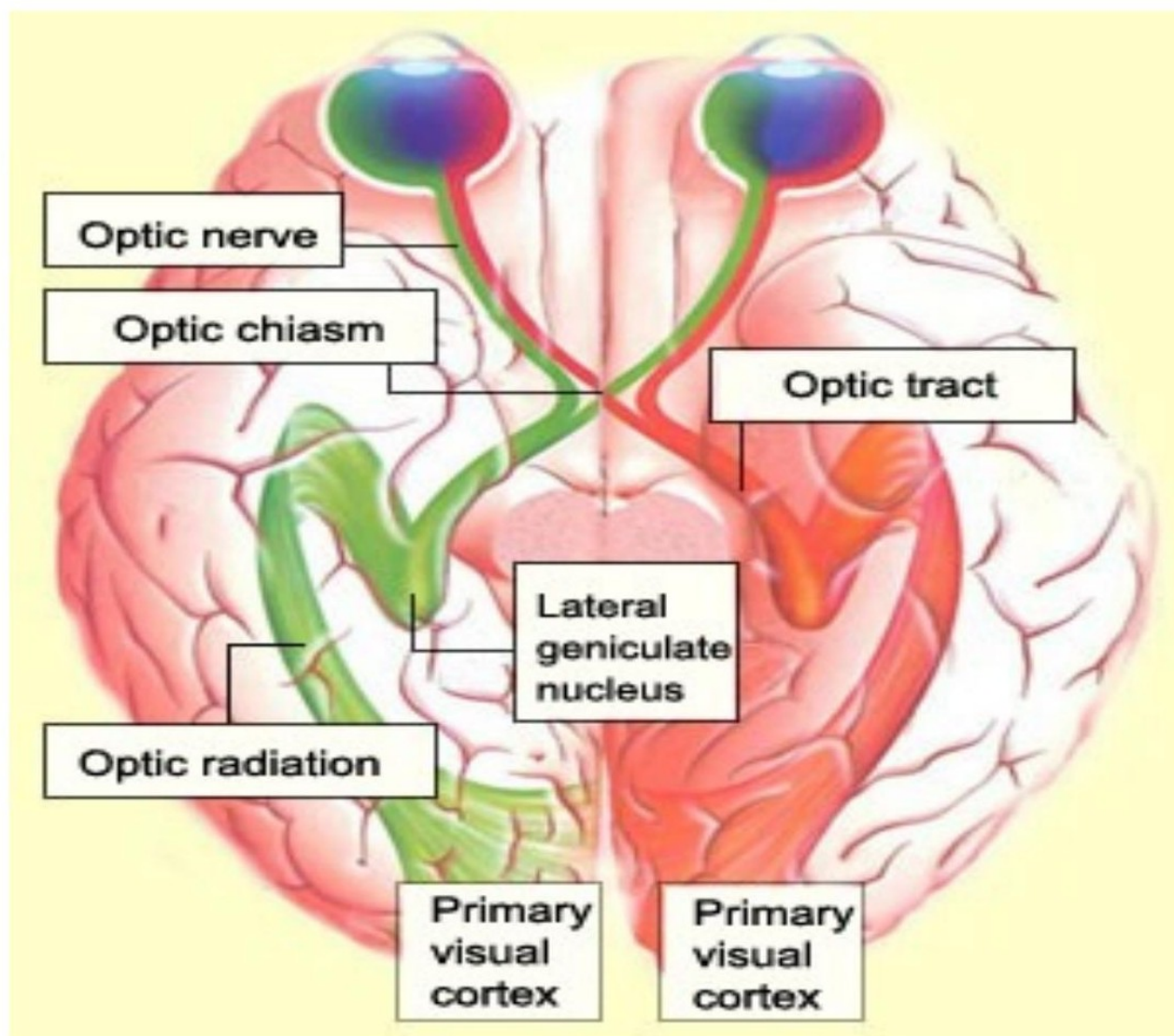
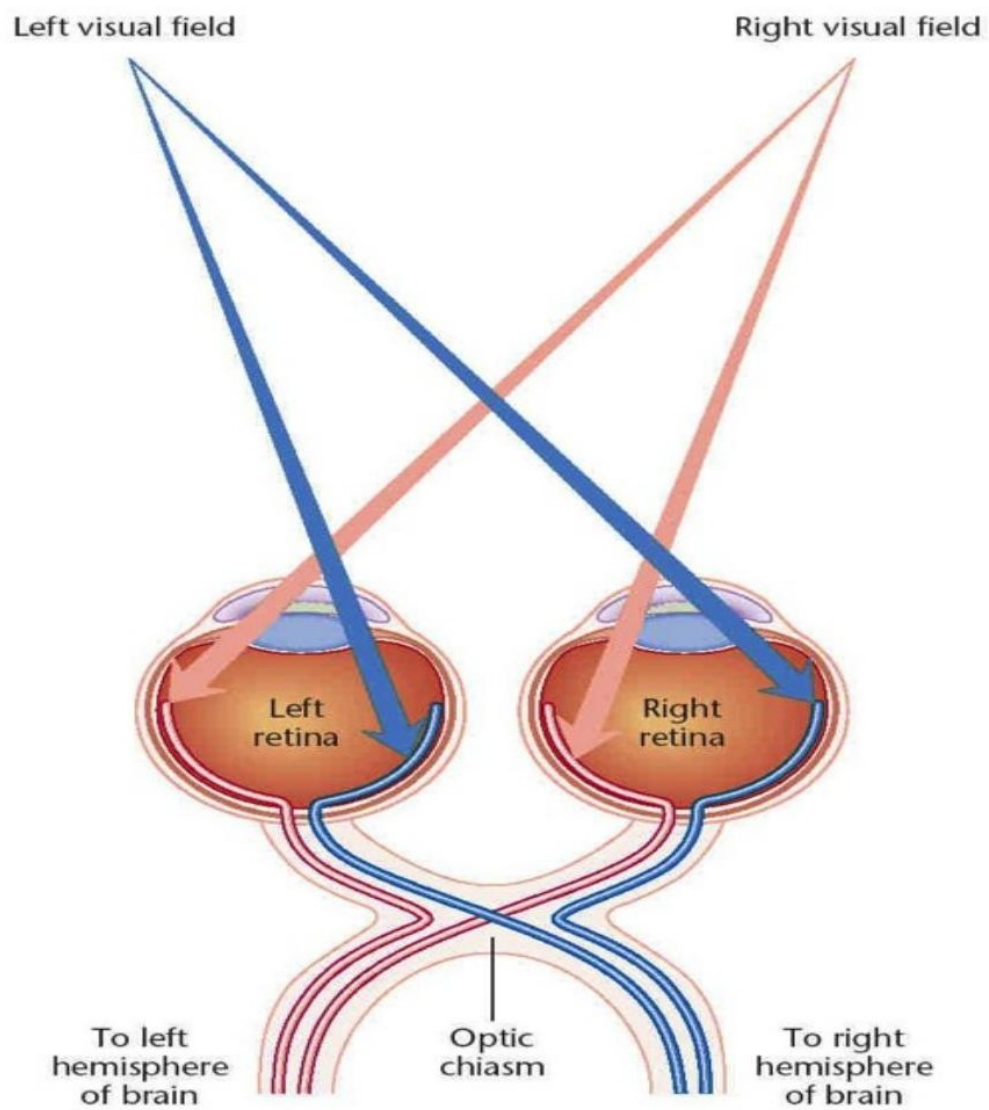


Figure L1.7 — Visual pathway anatomy from eye to cortex



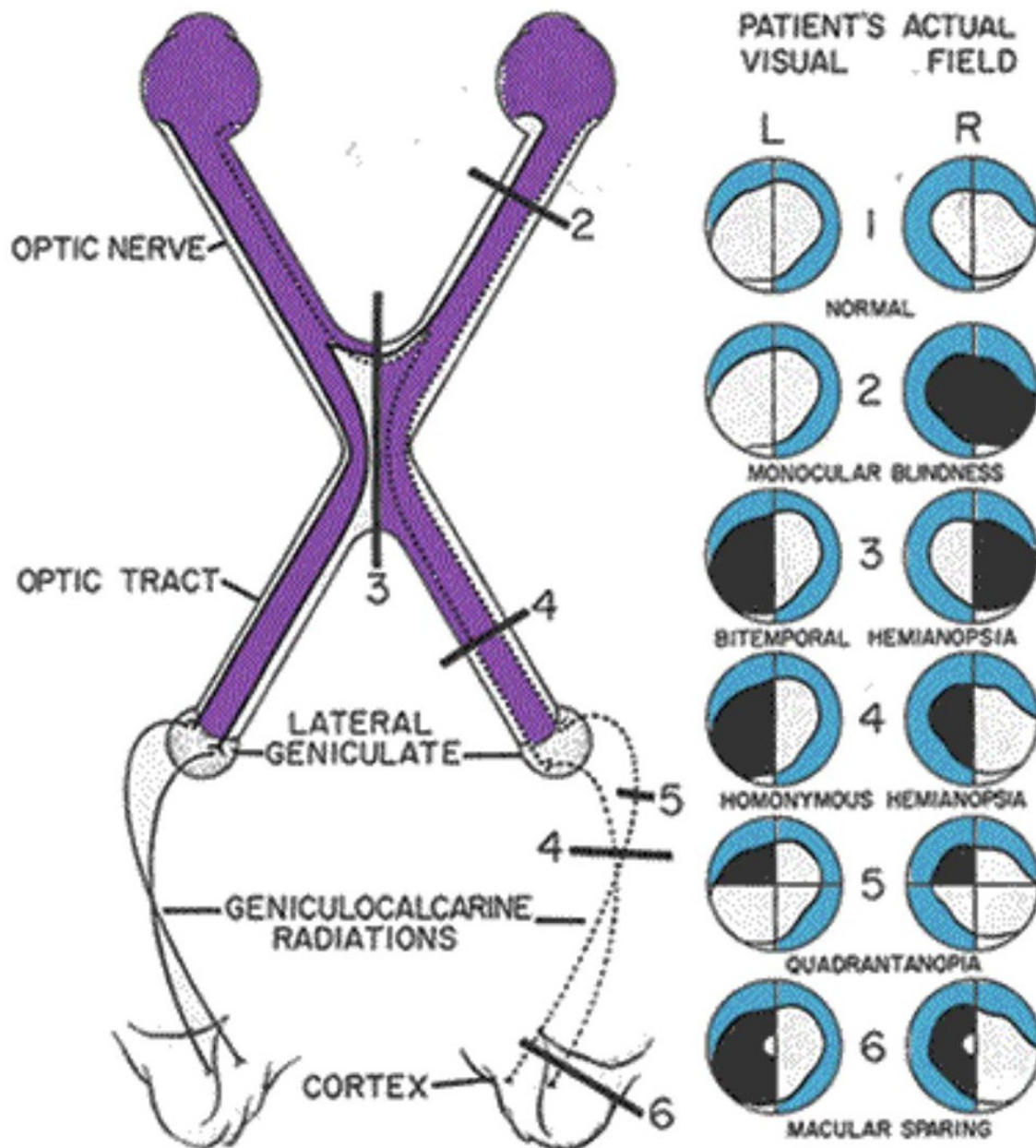
linical assessment.

Figure L1.8 — Optic chiasm: nasal retina crosses, temporal does not



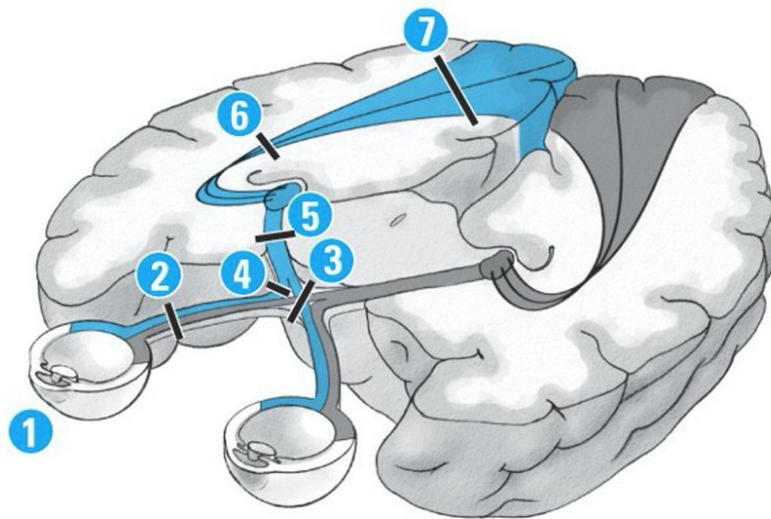
Figure L1.9 — Sagittal MRI: pituitary lesion compressing chiasm

Visual pathway anatomy



linical assessment.

Figure L1.10 — Homonymous hemianopia with macular sparing: cortical lesion



Patient's actual visual field

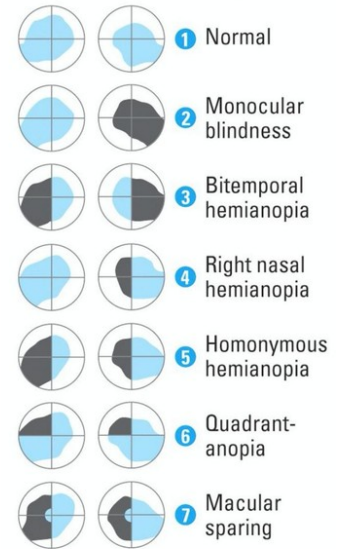
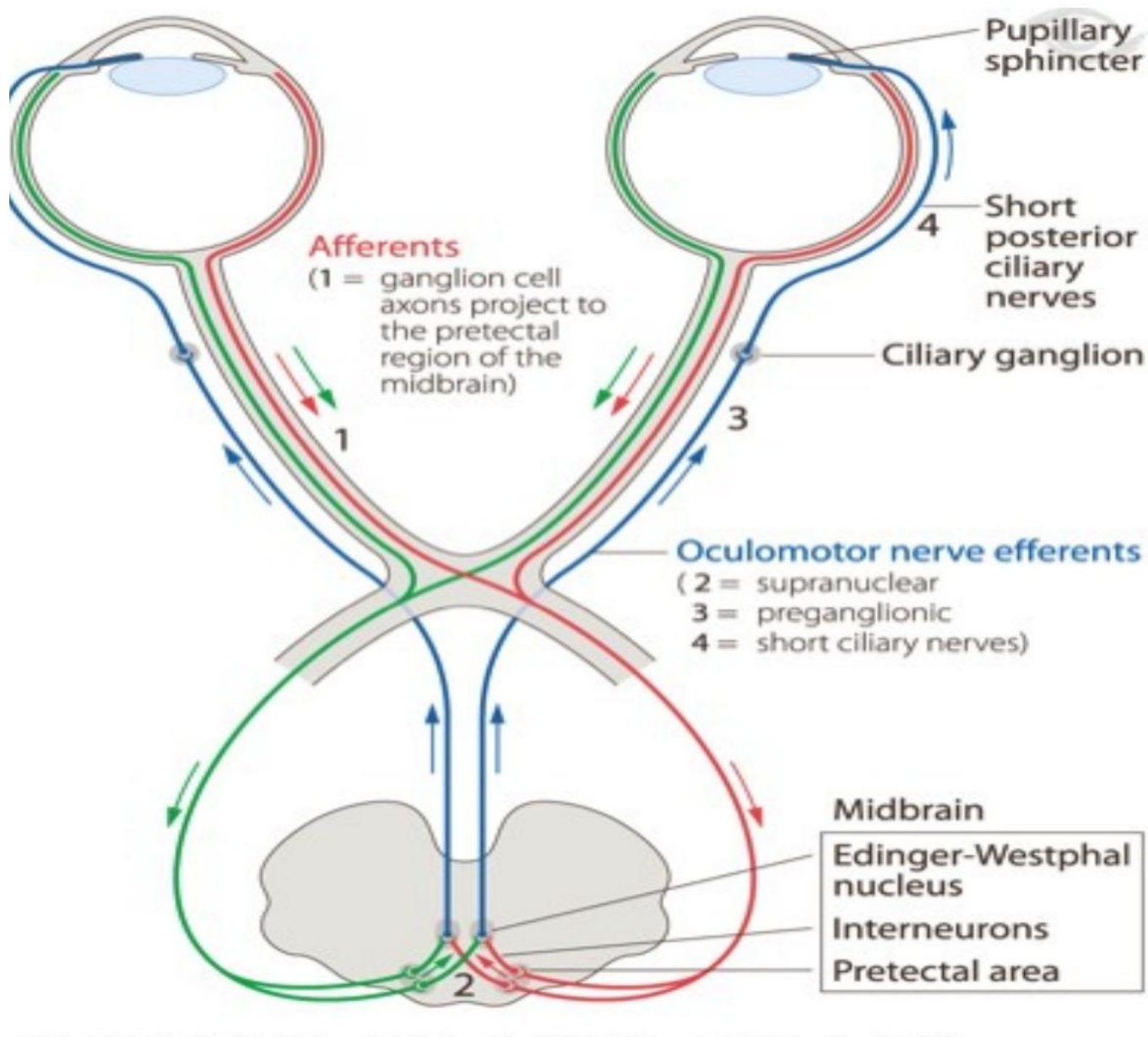
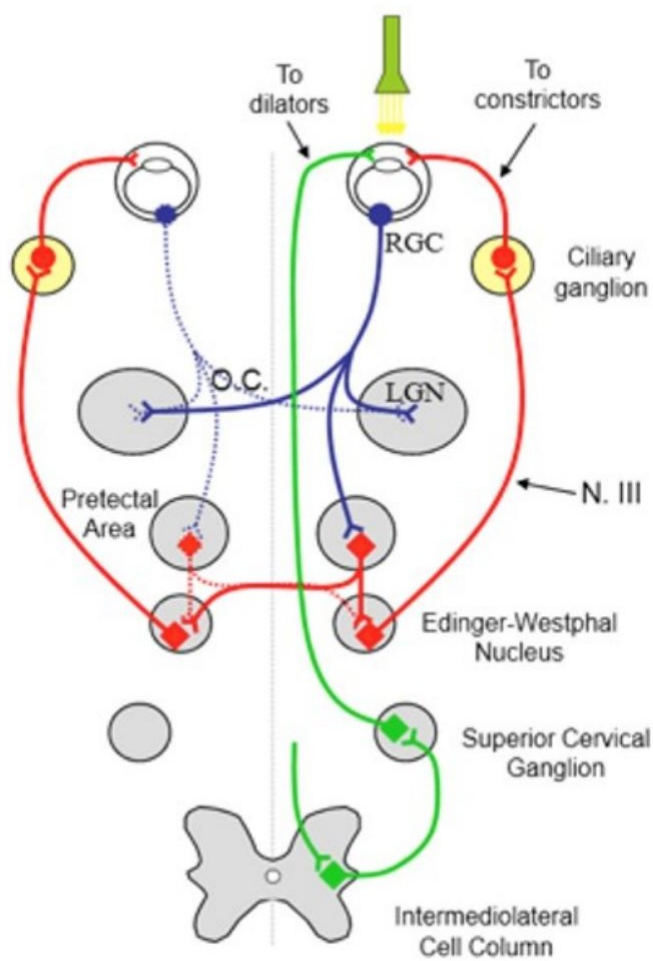


Figure L1.11 — Visual pathway lesions (1–7) and corresponding field defects



linical assessment.

Figure L1.12 — Pupillary reflex arc: afferent and efferent pathways



Normal pupillary reflex

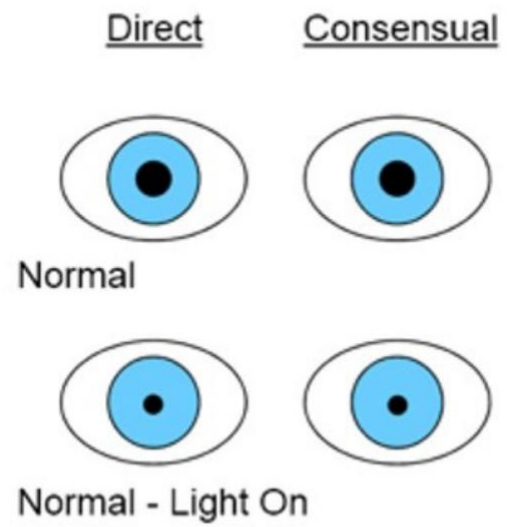


Figure L1.13 — Direct vs consensual light reflex

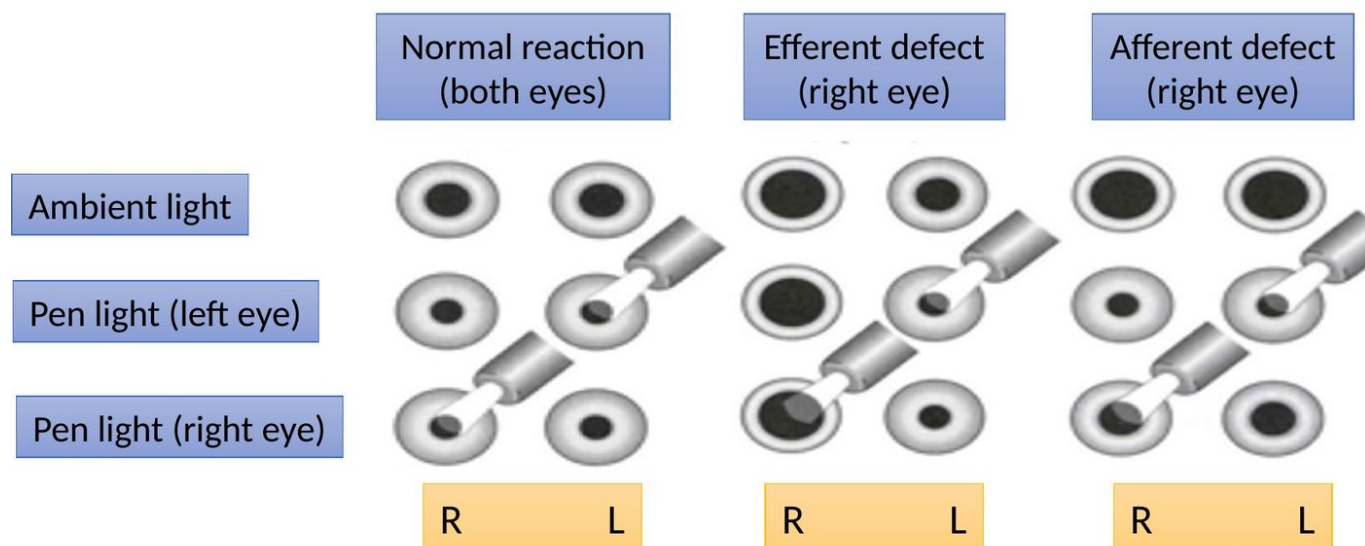
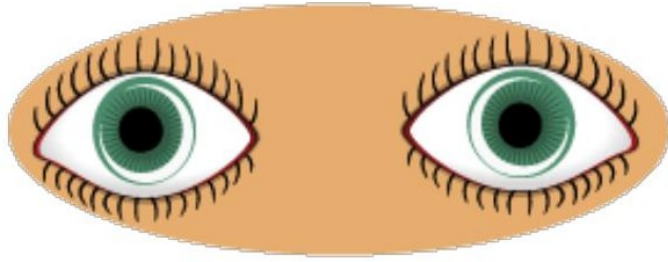
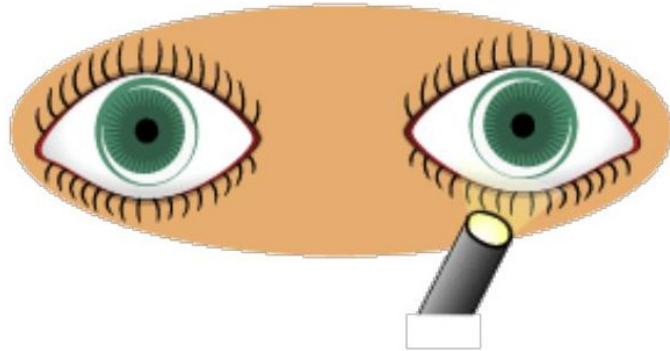


Figure L1.14 — Normal vs efferent vs afferent defect pupil responses

No Light



**Normal
Response
to Light**



**Positive
RAPD of
Right Eye**

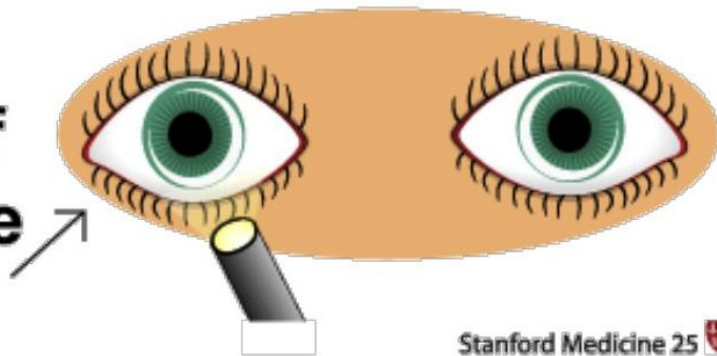


Figure L1.15 — Swinging torch test: paradoxical dilation in RAPD

Extraocular muscles

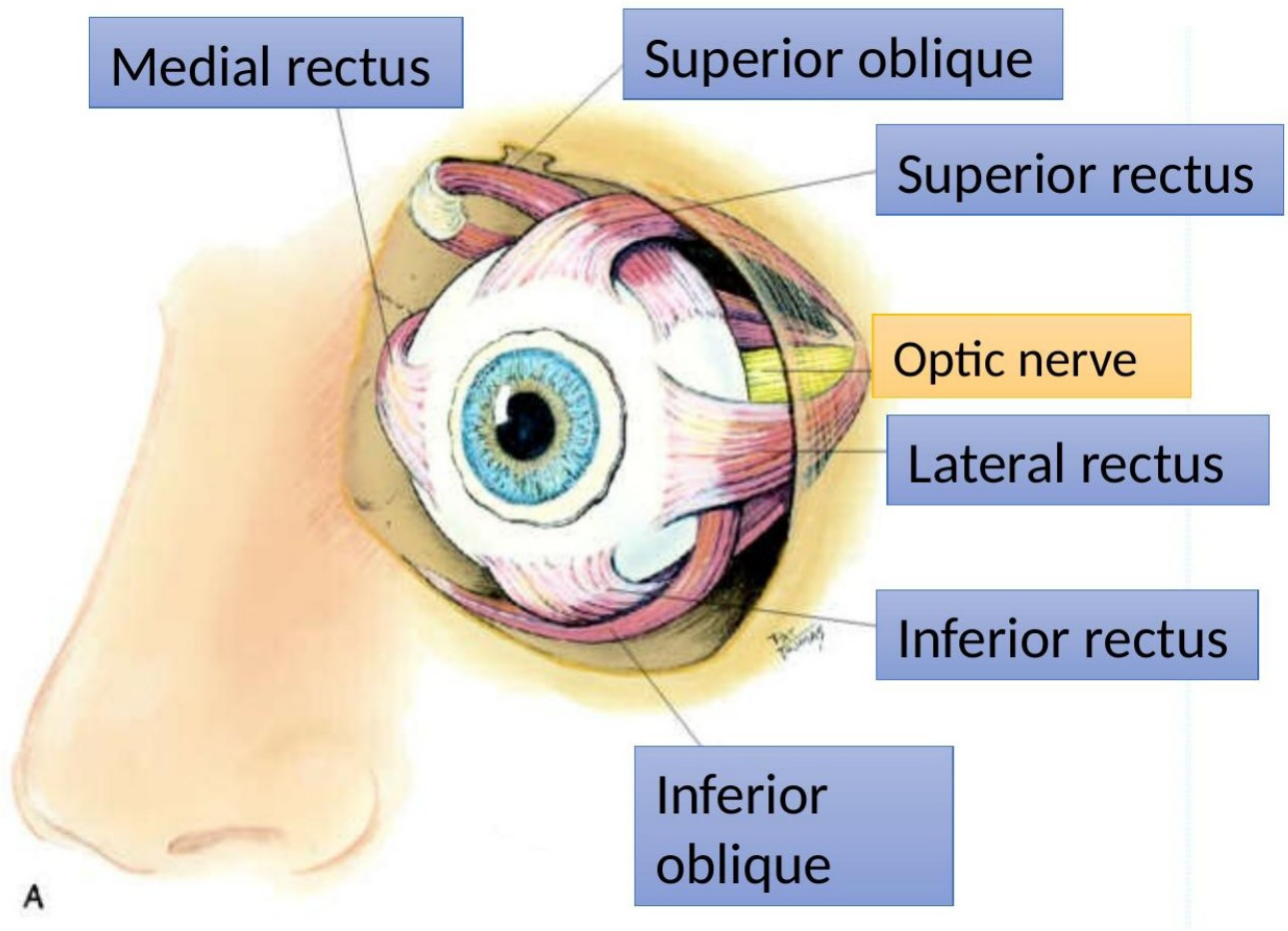


Figure L1.16 — Extraocular muscles of the right eye (anterior view)

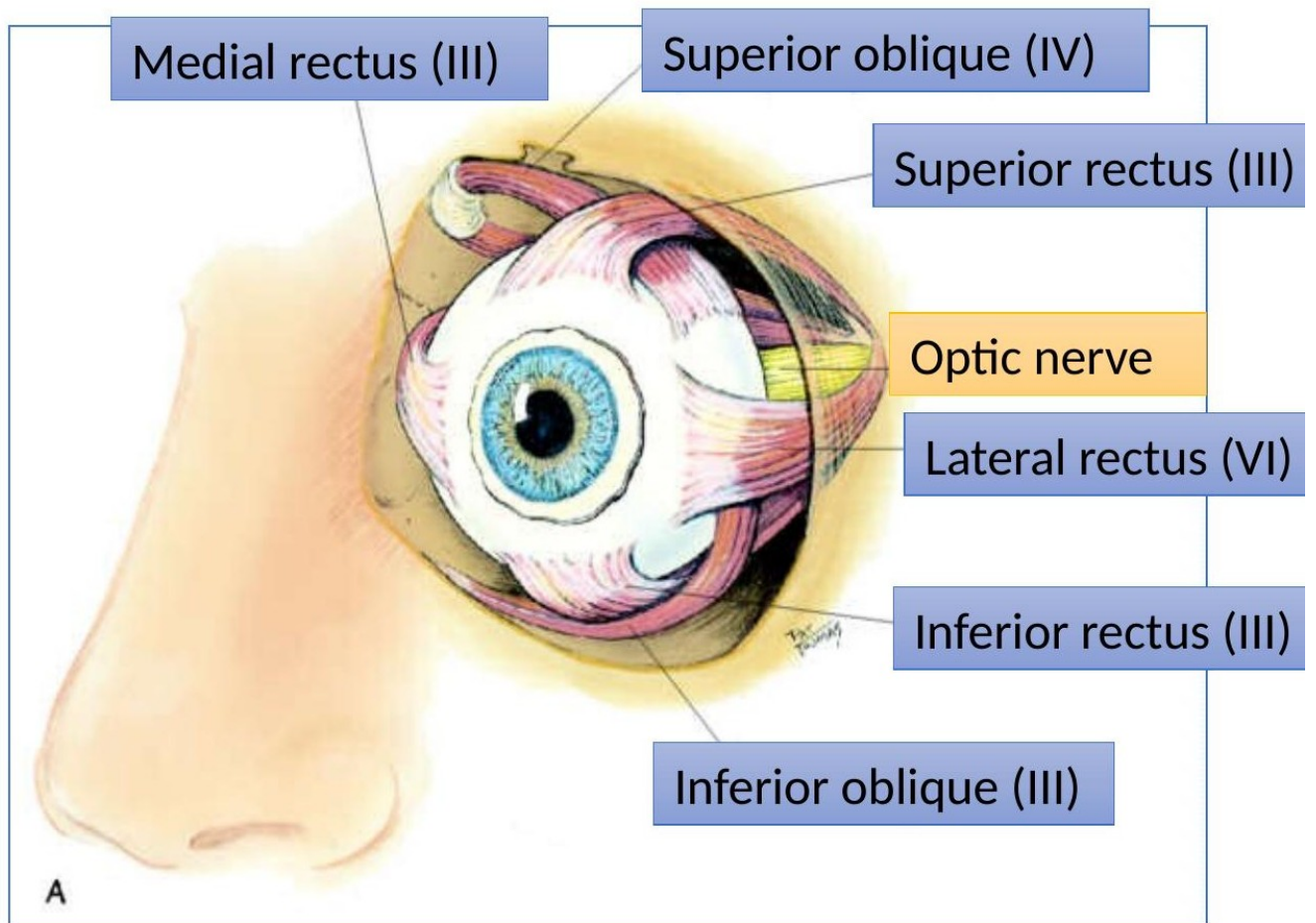
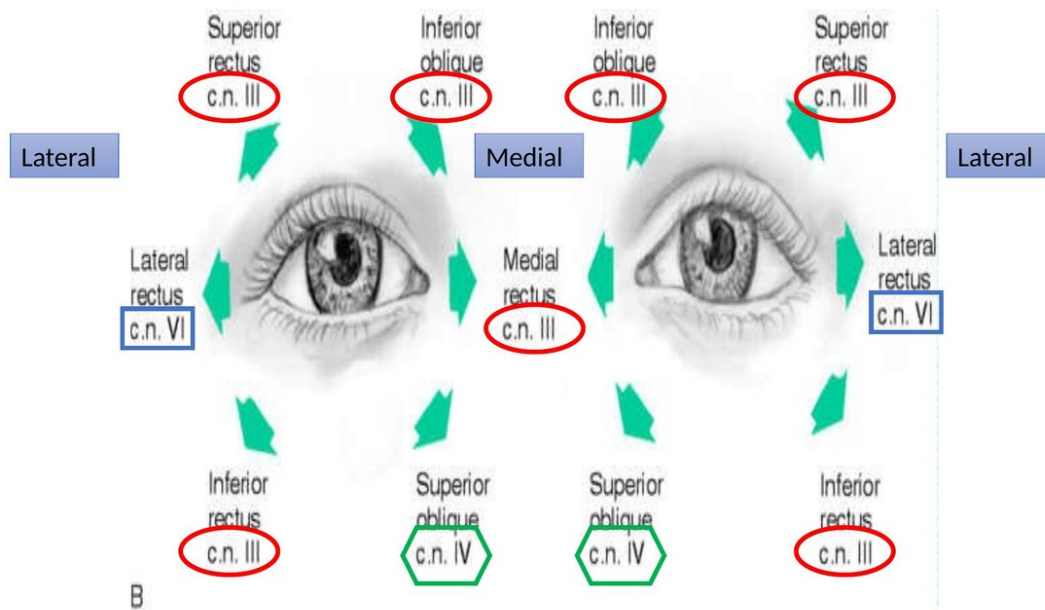


Figure L1.17 — Extraocular muscles with cranial nerve innervation



senses: Describe the structure and function of the visual, hearing and balance systems and explain their clinical assessment.

Figure L1.18 — Cardinal positions of gaze (H-test) with muscle and CN

III also innervates:

Somatic:

Levator palpebrae sup **X**

Parasympathetic:

Ciliary muscle (accommodation) **X**

Constrictor (sphincter) of the iris **X**

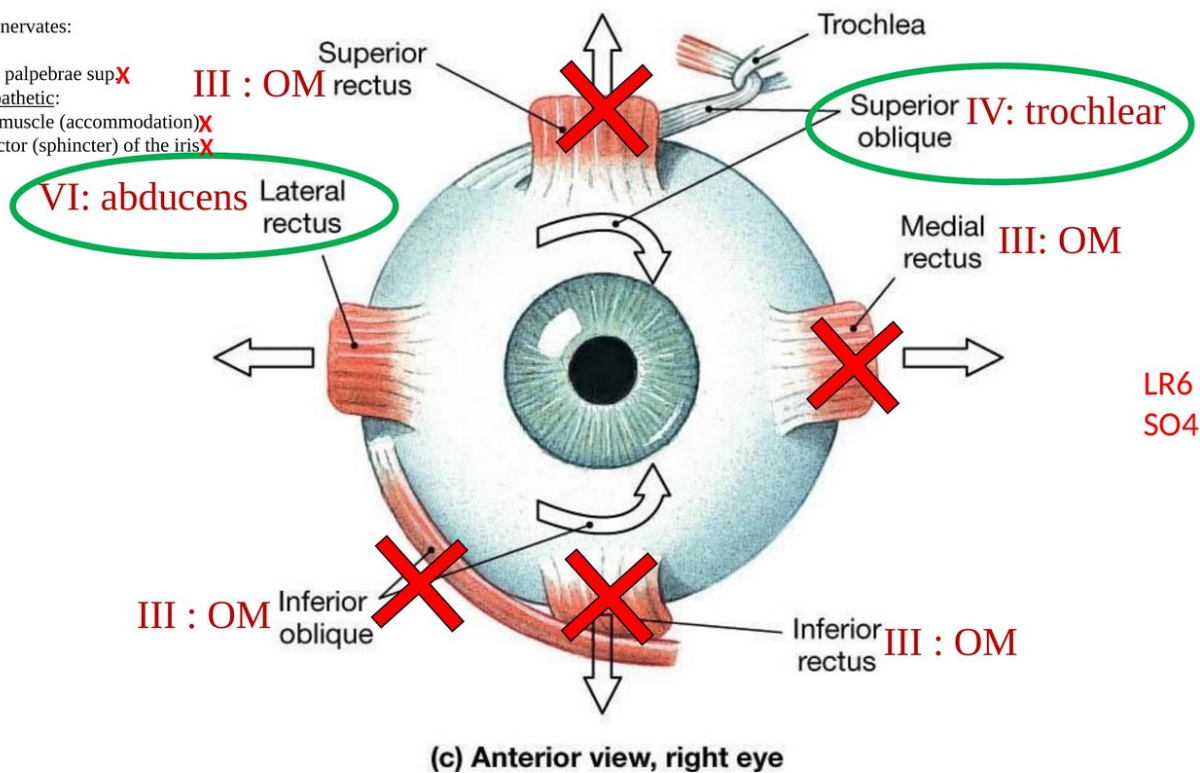


Figure L1.19 — Right 3rd nerve palsy: muscles knocked out (LR6 SO4 only working)

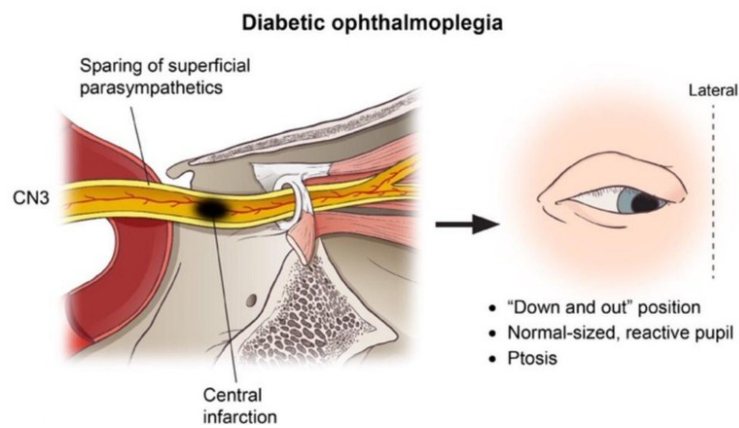
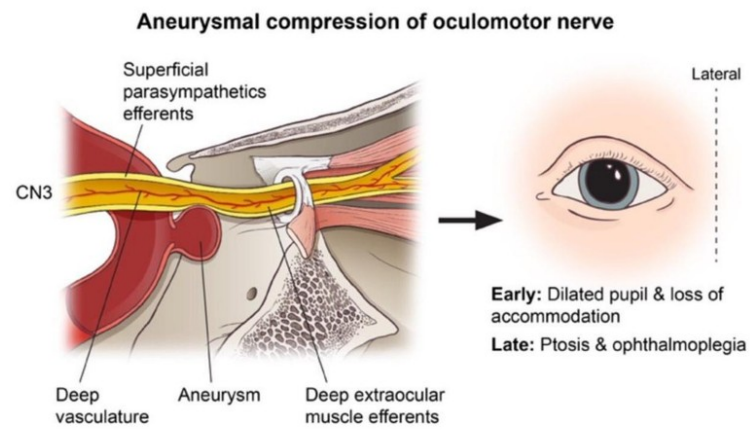


Figure L1.20 — Pupil-involving (surgical) vs pupil-sparing (medical) 3rd nerve palsy

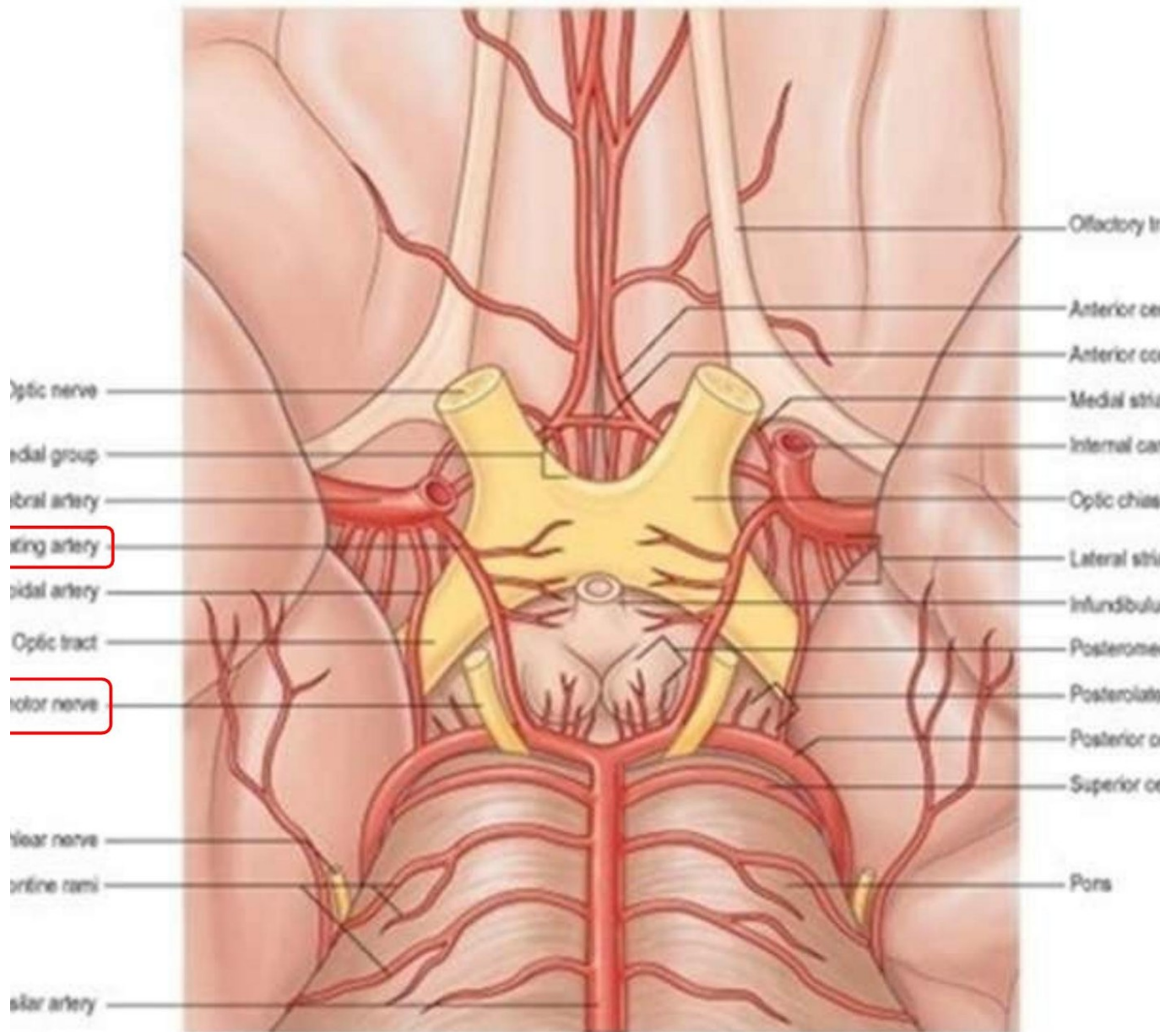


Figure L1.21 — Circle of Willis: PCom artery next to oculomotor nerve

THIRD **FOURTH** **SIXTH**



Describe the structure and function of the visual, hearing and balance systems and explain their clinical assessment.

Figure L1.22 — Clinical appearance: 3rd, 4th, 6th nerve palsy